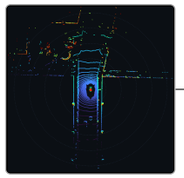
DSB perception: BEVFusion (detection + BEV segmentation) and FlashOCC (occupancy)

BEVFusion: Swin-T + LSS/DepthLSS (camera) and SparseEncoder (LiDAR) → ConvFuser → TransFusion / BEV-seg heads. FlashOCC: ResNet-50 + LSS view transform + 2D BEV encoder + channel-to-height occupancy head. Both consume the same 6 cameras; the image backbones differ (Swin-T vs ResNet-50).



LiDAR point cloud (BEV)

BEVFusion | camera + LiDAR

Voxelize
0.075 m (xy) · 0.2 m (z)

SparseEncoder
spconv

LiDAR BEV
256×180×180

Swin-T
+ GeneralizedLSSFPN

DepthLSS / LSS
view transform

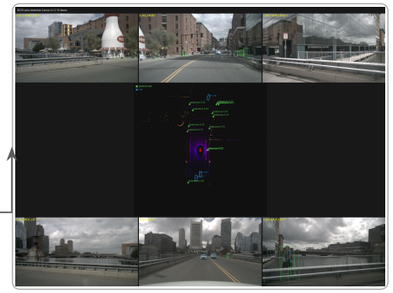
camera BEV
80×180×180

ConvFuser
80+256 → 256

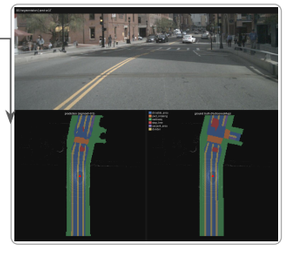
SECOND + SECONDFPN
512×180×180

TransFusion head
3D detection

BEV seg head
6-class map



detection: 6 cameras + centered BEV



BEV segmentation: pred vs GT

geometry

Extrinsics / aug
camera2lidar · lidar2ego
img_aug · lidar_aug



6 cameras (shared by both networks)

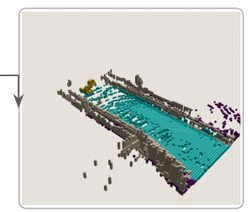
ResNet-50
+ CustomFPN

LSSViewTransformer
64×200×200

CustomResNet
+ FPN_LSS
256×200×200

BEVOCCHead2D
18 × 16 channels

FlashOCC | camera only



occupancy: 200×200×16

Grids: det 0.075×0.075×0.2 m voxel / 180×180 BEV; seg 0.1×0.1×0.2 m voxel / 128×128 BEV (+ 200×200 map @ 0.5 m); occ 0.4 m / 200×200×16 occupancy.